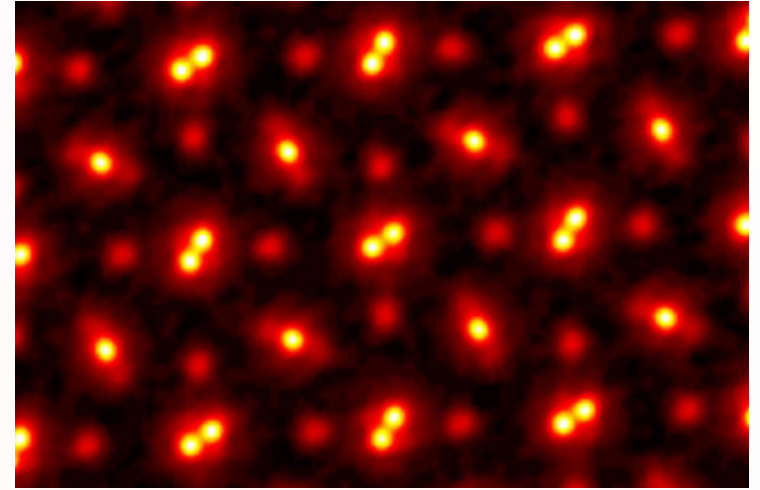




Phases of Matter

What is matter?

- Everything that has mass and takes up space
- Every material is a different type of matter
- The atom is the basic unit of matter.



Highest resolution image of atoms

What are atoms like?

- Atoms are always moving, vibrating, and bumping into each other.
- Atoms are so small that they do not reflect light, so you cannot see them with a normal microscope.
- A human hair is about a million times wider than an atom.
- Atoms themselves are mostly empty space. They have a dense core of protons and neutrons surrounded by a cloud of negatively charged electrons.
- Differences in the numbers of positively charged protons make different kinds of atoms.

What kinds of matter are there?

- *There are 94 different kinds of atoms found in nature.*
- *Each different kind of atom is called an **element***
- *Differnt atoms form bonds to make **molecules**, which may have many different elements.*
- *Most materials we find in nature are mixtures of different molecules.*

Periodic Table of the Elements

1

1A

H

Hydrogen

1.008

2

2A

He

Helium

4.002602

Atomic Number →

1

← Symbol

← Atomic Weight

1.008

← Name

Hydrogen

3	Li	Be											13	B	C	N	O	F	Ne	
4	Li	Be											14	B	C	N	O	F	Ne	
5	Na	Mg											15	Al	Si	P	S	Cl	Ar	
6	Na	Mg											16	Al	Si	P	S	Cl	Ar	
7	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr		
8	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr		
9	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe		
10	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe		
11	Cs	Ba	57-71 Lanthanoids		72	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
12	Cs	Ba	57-71 Lanthanoids		72	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
13	Fr	Ra	89-103 Actinoids		104	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Nh	Fl	Mc	Lv	Ts	Og
14	Fr	Ra	89-103 Actinoids		104	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Nh	Fl	Mc	Lv	Ts	Og
15	La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu					
16	La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu					
17	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr					
18	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr					



Properties of matter

- Matter has *physical* and *chemical* properties
- *Physical properties* include everything we can observe without changing the identity of the material

Examples include:

Observable Properties

- *color*
- *mass*
- *volume*
- *luster (shininess)*

Mechanical Properties

- *hardness (how hard it is to scratch)*
- *brittleness (how easily it shatters)*
- *malleability (how easily it bends)*
- *ductility (how easily it can be pulled into a wire)*
- *tensile strength (how hard it is to tear)*

Other Properties

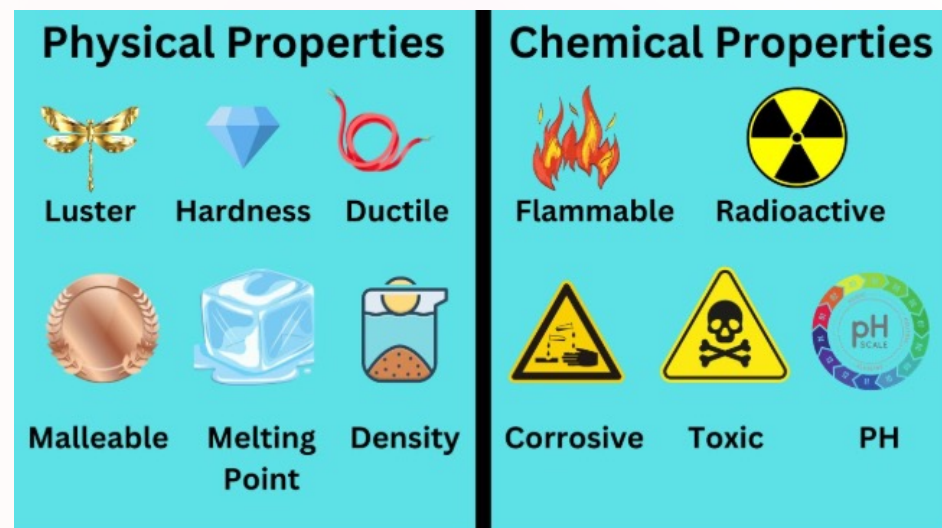
- *electrical conductivity*
- *magnetic properties*
- *melting point*
- *boiling point*

Properties of matter

- Matter also has chemical properties, which can only be observed when the atoms rearrange themselves in new ways, often making new molecules.
- When this happens, a new substance is formed and the change is a **chemical change**.

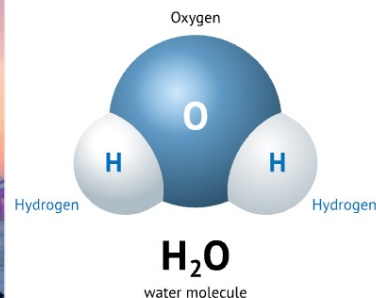
Examples of chemical properties include:

- flammability (the ability to burn)
- reactivity (how a substance interacts with other chemicals)
- toxicity (harmfulness to living organisms)
- oxidation state (a measure of electrons an atom can gain or lose)
- heat of combustion (energy released when a substance burns)
- chemical stability (resistance to chemical change)



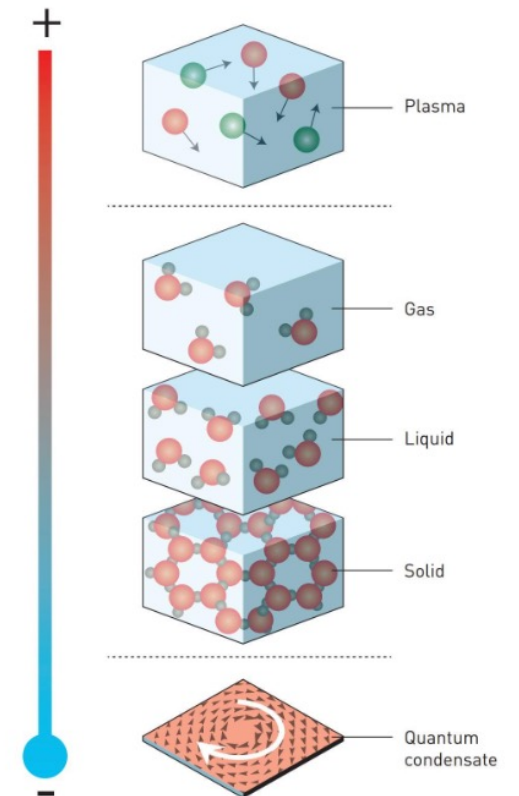
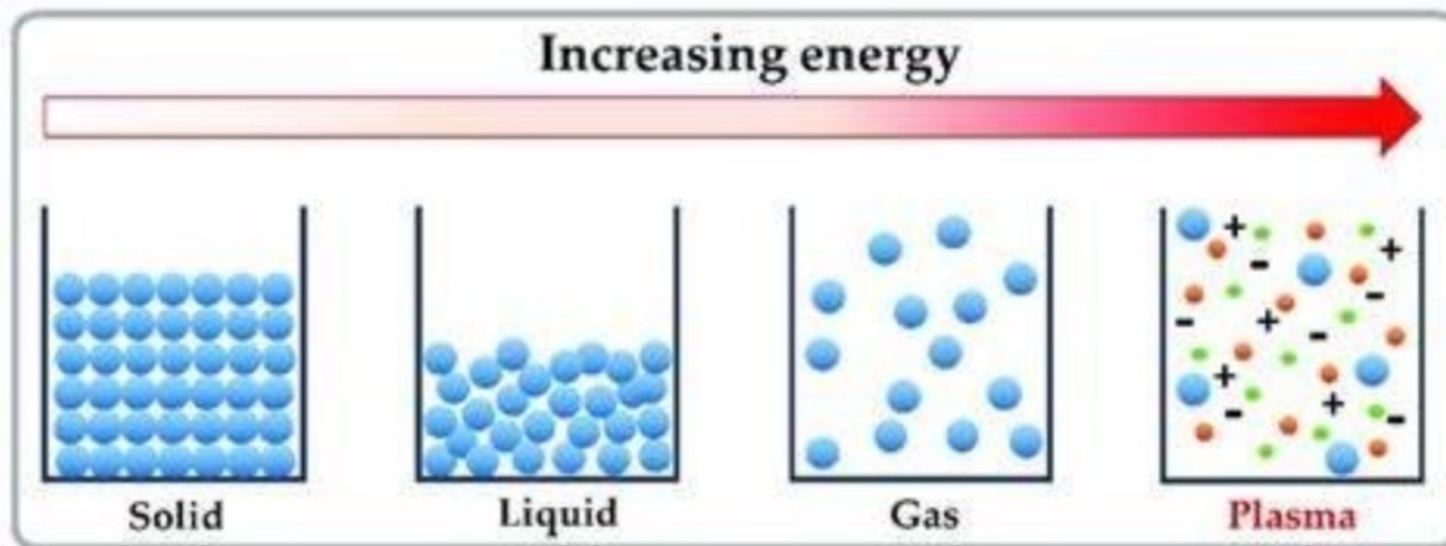
States of matter

- We rarely find one atom by itself, except for in the mostly empty space between stars.
- When atoms or molecules (structures of atoms that are bonded together) come together, they can form different states of matter
- A gold atom by itself is not golden or shiny. A water molecule by itself is not wet.
- Physical properties are the result of many atoms or molecules interacting with each other.
- Each state of matter has different physical properties. For example, water ice is quite hard, but liquid water will allow dense objects to pass through. Water vapor can burn your skin. All forms of water have the same chemical identity: H_2O



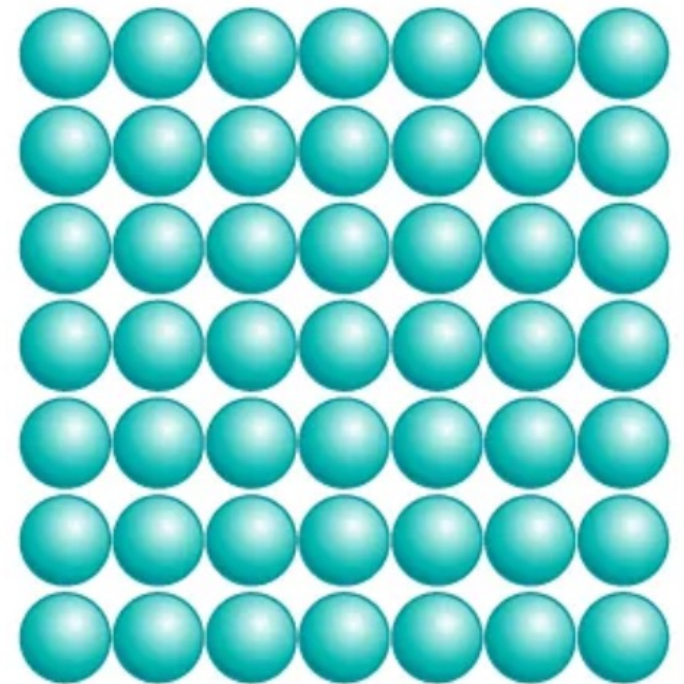
How many states of matter are there?

- The three main states of matter we interact with every day are **solid, liquid and gas**.
- We see **plasma** in the form of lighting on earth, and in stars like our Sun.
- There are more than 12 states of matter scientists have identified. The newly discovered "exotic" states of matter, like the **Bose-Einstein condensate** and **Fermionic condensate** only occur at extremely low temperatures in laboratories.

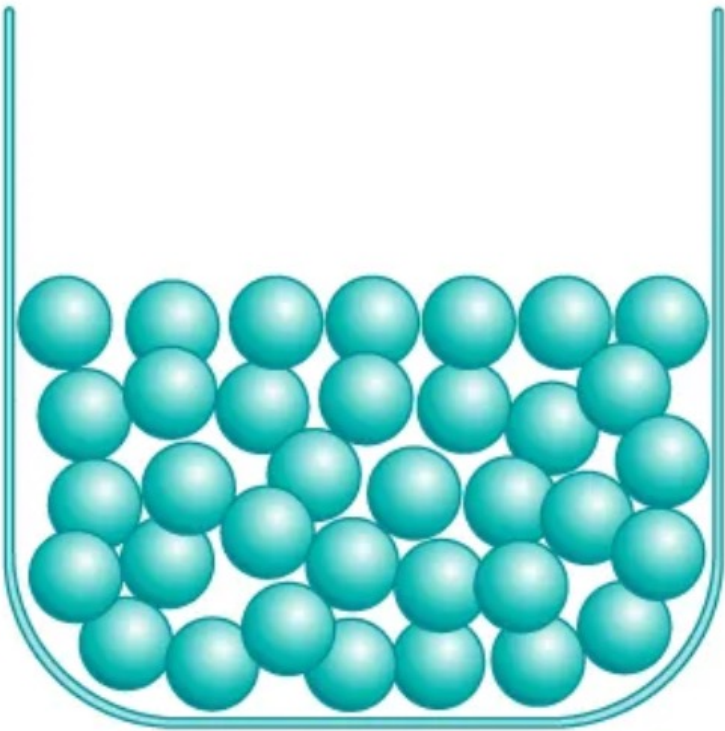


Solids

- Solids have a *fixed shape and fixed volume*.
- Here, "fixed" means it does not change.
- The particles in a solid vibrate with heat energy
- The particles in a solid do not have enough energy to move away from each other.



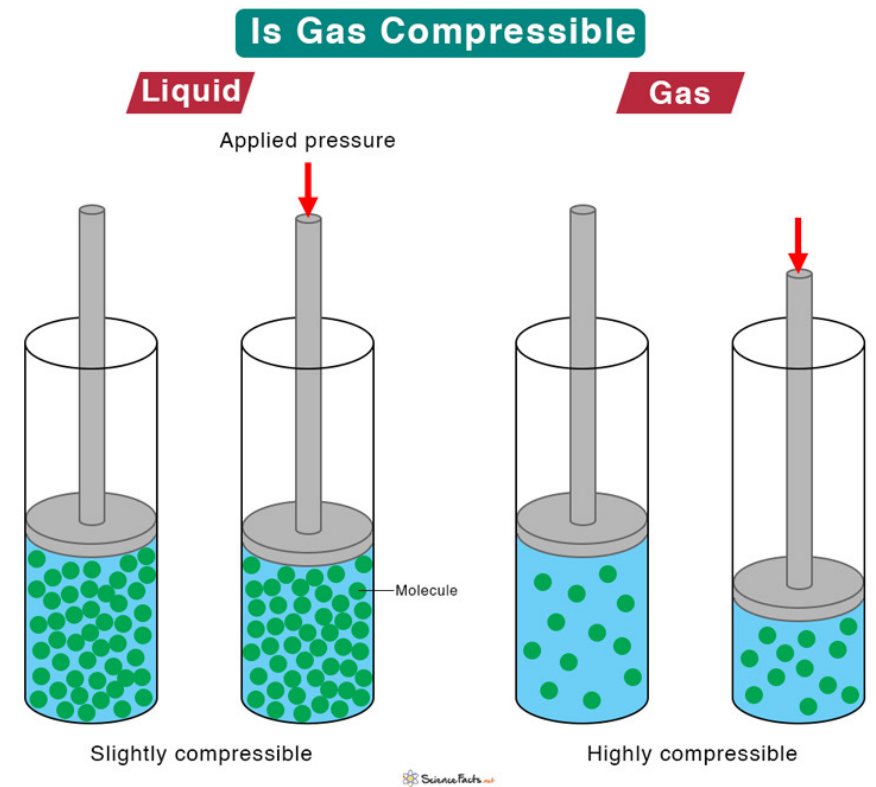
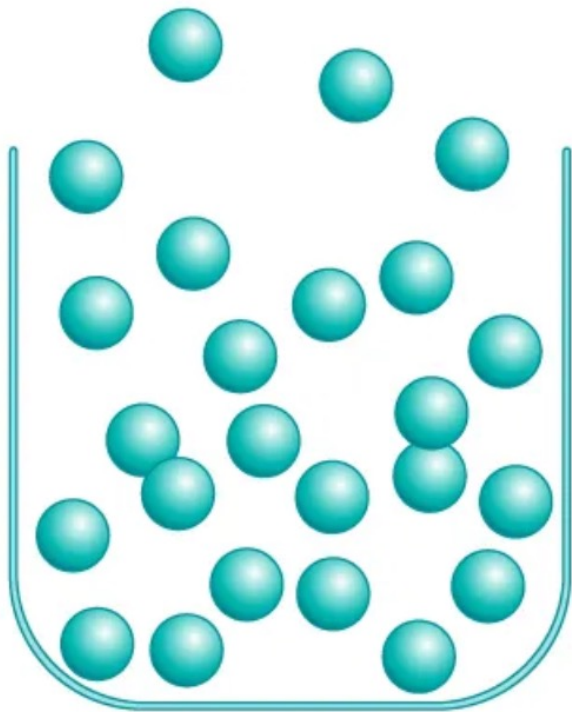
Liquids



- Liquids have a **fixed volume, but no fixed shape**
- Liquids take the shape of their container
- The particles in a liquid have enough energy to move past one another
- The attraction between particles is still strong enough that the particles touch.

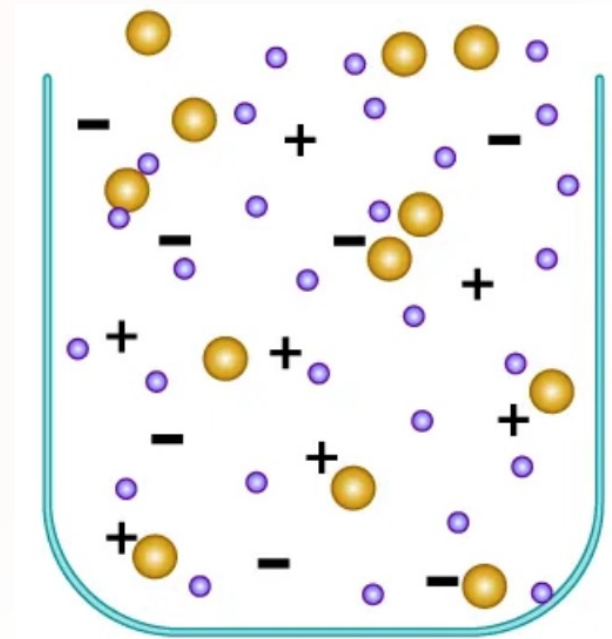
Gasses

- A gas has **no fixed shape and no fixed volume**.
- No fixed volume means a gas is **compressible**. It can be squashed.
- The particles in a gas have enough energy to move around independently



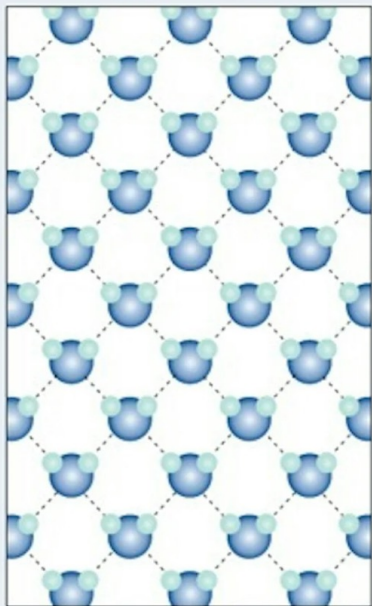
Plasma

- Stars and lightning are examples of plasma
- In a plasma, particles are heated so much that the atoms lose their electrons and become charged
- Charged particles behave differently than normal gas particles, so plasma is often called "the fourth state of matter".
- Charged particles are called **ions**, which is why changing into a plasma is called **ionization**.
- The reverse process is called **recombination**.

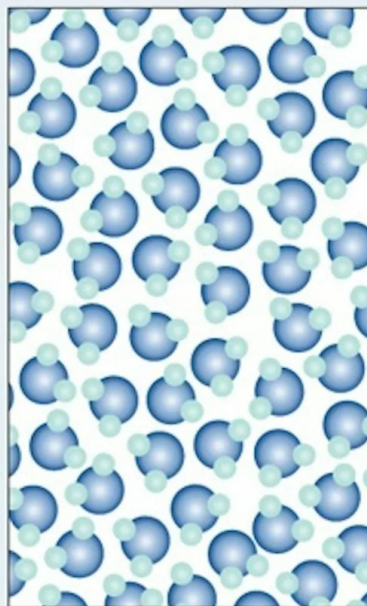


Phase Changes

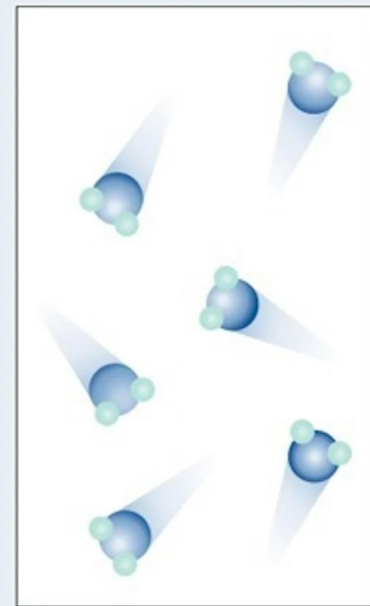
- All phase changes are **physical changes**
- The identity of the substance does not change, only the arrangement of particles.
- Changes in thermal energy, or heat, cause phase changes
- Changes in thermal energy, or heat, cause phase changes
- Substances heat up when they experience increased pressure, so pressure is also important when considering at what temperature a phase change will occur.



Water molecules in solid ice.



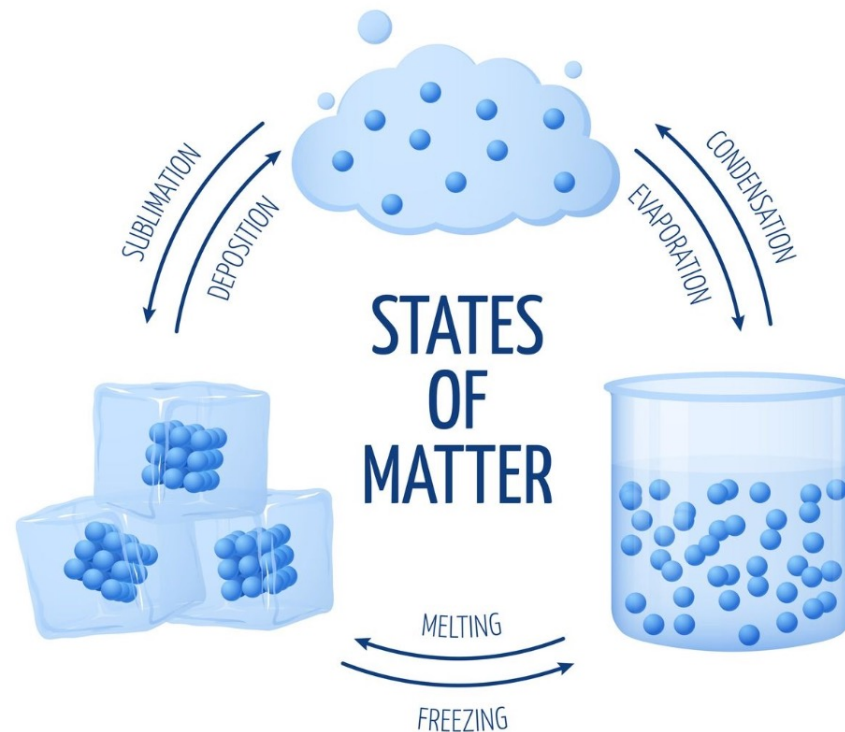
Water molecules in liquid water.



Water molecules in water vapour
– a gas.

Types of Phase Changes

- Phase changes are also called changes of state. "phase of matter" is another way to say "state of matter".
- As heat is absorbed (endothermic) a substance can condense, freeze, or deposit.
- As heat is released (exothermic) a substance can melt, vaporize, or sublimate



Phase change and temperature

- *As a substance changes state, its energy remains constant (does not change) until the change of state is complete.*
- *If water is heated to 100 at 1 atmosphere of pressure, the temperature will stay at 100 even as more heat is added. The extra energy goes to breaking the attraction between the water molecules*
- *When a substance is hotter than it could normally be without changing state, it is called **superheated**.*
- *When a substance is colder than it could normally be without changing states, it is called **supercooled**.*